

# TEWODROS HAILEGEBEREL

## DIGITAL TRANSFORMATION, ENTERPRISE GIS SYSTEMS LEADER

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### PROFESSIONAL SUMMARY

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Senior digital transformation and enterprise systems leader with 20+ years of experience helping governments, utilities, development organizations, and technology teams design, implement, govern, and sustain complex digital platforms across Africa and the United States.

- Led technical and program oversight for USAID/Ethiopia's approximately \$63 million national Digital Health Activity.
- Deep experience spanning enterprise GIS, digital health architecture, interoperability, data governance, system integration, analytics, technical assistance, and institutional capacity strengthening.
- Built and managed a 40+ person technology organization and worked across government leadership, operational teams, engineers, developers, donors, implementing partners, vendors, and end users.
- Combines executive-level advisory experience with hands-on understanding of ArcGIS Enterprise, enterprise geodatabases, APIs, Python, SQL, dashboards, ETL, interoperability, cloud platforms, and emerging AI tools.

### CORE LEADERSHIP CAPABILITIES

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Digital Transformation Strategy • Technical Assistance • Enterprise Systems Architecture • Enterprise GIS • Digital Health Architecture • Systems Integration • Interoperability • Data Governance • Technical Product Ownership • Requirements Engineering • Implementation Planning • Quality Assurance • Production Readiness • Risk Management • Government & Partner Engagement • Knowledge Management • Capacity Strengthening • Institutional Handover • Program & Portfolio Oversight • Team Leadership • Executive Decision Support

### TECHNICAL EXPERTISE

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**Enterprise GIS & Data Systems:** ArcGIS Enterprise • ArcGIS Pro • QGIS • Enterprise Geodatabases • PostgreSQL/PostGIS • SQL • Python • GIS Data Governance • Web GIS • APIs • ETL • Enterprise Data Management • Spatial Analytics • Dashboard Development

**Digital Health & Interoperability:** DHIS2 • eCHIS • EMR • eLMIS • iHRIS • OpenHIM • HL7 FHIR • Health Information Exchange • APIs • Interoperability Architecture • Data Pipelines • Digital Health Governance • Health Information Architecture

**Enterprise & Utility Integration:** GIS-ERP/SAP Integration • SCADA • ADMS/OMS • MDMS • Asset Systems • Customer Systems • Enterprise Architecture • Operational Data Integration

**ArcGIS Utility Network:** Working knowledge of Utility Network architecture, topology, validation, tracing, branch-versioned editing concepts, migration readiness, integration dependencies, and operational utility workflows.

**Analytics & Emerging Technology:** Power BI • Tableau • Azure • AWS • Google Apps Script • Generative AI • Gemini API Prototyping • AI-Enabled Analytics • Workflow Automation

### PROFESSIONAL EXPERIENCE

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#### Senior GIS Engineer

*Adam Smith International | Public-Sector & Utility Advisory | Addis Ababa, Ethiopia | Dec 2025 – Mar 2026*

- Advise Ethiopian public-sector and utility leaders on enterprise GIS modernization, digital systems, data governance, implementation planning, and alignment between technology investments and operational requirements.
- Develop and review enterprise GIS strategies, technical architectures, governance arrangements, implementation roadmaps, requirements, procurement inputs, and acceptance criteria.
- Review ArcGIS Enterprise and Utility Network requirements and advise stakeholders on architecture considerations, data readiness, governance, integration dependencies, operational workflows, and implementation risks.
- Define integration approaches linking GIS with SAP/ERP, SCADA, ADMS/OMS, MDMS, asset, customer, and operational systems to reduce information silos and strengthen enterprise-wide data use.
- Strengthen enterprise GIS supportability through data standards, QA/QC procedures, validation requirements, ownership models, publishing processes, and structured asset-data workflows.
- Coordinate technical activity across GIS, infrastructure, databases, operational teams, vendors, consultants, development partners, and executive stakeholders.
- Assess production-readiness requirements including platform governance, data quality, administration responsibilities, integration dependencies, knowledge transfer, and long-term institutional ownership.
- Produce technical reports, architecture recommendations, maps, dashboards, procurement documentation, and decision-support materials for senior leadership and implementation teams.
- Provide technical guidance and knowledge transfer to strengthen client capability to administer, operate, and sustain enterprise GIS environments.

## **Regional GIS & Digital Health Consultant**

*Palladium | Data.FI Project | Burundi / Remote | Oct 2025 – Dec 2025*

- Advised Burundi's Ministry of Health and national technical partners on digital health architecture, interoperability, data governance, analytics, and sustainable integration of national health information systems.
- Assessed workflows and system relationships across SIDAInfo, IBIPIMO, eLMIS, DHIS2, and OpenHIM, translating findings into integration requirements, architecture decisions, implementation actions, and technical documentation.
- Co-developed interface control documentation and functional requirements for OpenHIM-based interoperability, defining data exchanges, system responsibilities, technical dependencies, and standards-based approaches including HL7 FHIR where appropriate.
- Supported design of integrated data pipelines and synchronization approaches linking clinical, laboratory, logistics, and national reporting systems.
- Worked with government, program, and technical stakeholders to clarify data flows, system ownership, integration responsibilities, readiness requirements, and implementation sequencing.
- Translated program requirements and operational needs into technical specifications and implementation recommendations that software, interoperability, and Ministry teams could use to move from architecture discussions into implementation.
- Tracked architecture decisions, system readiness, technical risks, dependencies, responsibilities, and implementation actions to strengthen follow-through across participating teams.
- Led GIS, dashboard, and analytical work integrating program and geographic information to identify service gaps and support national and facility-level decisions.
- Produced technical guidance, architecture documentation, decision-support materials, and stakeholder briefings translating complex technical issues into actionable recommendations.
- Supported institutional capacity strengthening through knowledge transfer on systems architecture, interoperability, data governance, analytics, and long-term government ownership.

## **Senior Digital Health Advisor**

*USAID/Ethiopia Mission, Health Office | Addis Ababa, Ethiopia | Oct 2022 - Jul 2025*

- Served as senior advisor and principal technical oversight lead for USAID/Ethiopia's approximately \$63 million Digital Health Activity, providing technical, programmatic, performance, and contract oversight.
- Provided technical leadership for national digital health investments, translating Ministry of Health priorities, frontline requirements, user needs, and implementation constraints into system requirements, technical guidance, rollout priorities, and program decisions.
- Reviewed digital health architecture, interoperability approaches, infrastructure requirements, hosting models, data governance, security considerations, implementation plans, and system readiness across national platforms.
- Supported national systems spanning eCHIS, EMR, DHIS2, eLMIS, health workforce platforms, pharmaceutical regulatory systems, equipment management, facility registries, health information exchange, and analytics.
- Worked with Ministry of Health leaders, implementing partners, software teams, vendors, universities, donors, and local technology organizations to align digital solutions with national health information architecture, data-governance requirements, interoperability standards, and government priorities.
- Translated field requirements, operational workflows, stakeholder feedback, and government needs into functional requirements, technical recommendations, procurement inputs, implementation priorities, and acceptance criteria.
- Supported interoperability and data-exchange planning through APIs, shared data standards, health information exchange, and integration approaches designed to reduce fragmentation between national platforms.
- Reviewed software and implementation deliverables, testing approaches, deployment readiness, implementation risks, performance indicators, and corrective actions to strengthen quality during national rollout and scale-up.
- Supported structured user-feedback and testing loops connecting frontline users, government stakeholders, implementing partners, and technical teams around system usability, functionality, and implementation challenges.
- Reviewed annual work plans, budgets, procurement packages, performance indicators, technical deliverables, risks, and donor reporting and recommended corrective actions and revised priorities where required.
- Used dashboards, implementation data, research, system-performance information, and stakeholder feedback to identify implementation gaps and support senior government and USAID decision-making.
- Supported digital initiatives reaching 8,000+ health posts and 1,300+ health facilities, alongside national workforce, logistics, regulatory, facility-data, and equipment-management platforms.
- Developed and reviewed technical notes, decision papers, architecture recommendations, executive briefs, implementation guidance, presentations, and knowledge products for senior government, donor, and program leadership.
- Supported institutional handover and sustainability through governance arrangements, SOPs, workforce development, technical guidance, training, knowledge transfer, and clearly defined government ownership responsibilities.
- Supported AI-enabled health analytics and digital innovation with academic and local partners, assessing emerging technologies for potential value, governance requirements, infrastructure readiness, maintainability, and implementation risk.
- Supported 15+ youth-led digital service enterprises involving more than 100 young professionals and approximately 400 health facilities, increasing local capability to maintain and support digital systems.

## **Geospatial & Knowledge Management Specialist**

*USAID/Ethiopia Mission, Program Office | Addis Ababa, Ethiopia | May 2021 - Oct 2022*

- Served as a Mission-level technical resource for enterprise GIS, data management, knowledge management, analytics, and decision-support capabilities spanning health, humanitarian, resilience, infrastructure, and development programs.
- Advised Mission leadership and technical offices on GIS strategy, enterprise data, reporting, information management, analytical methods, and technology approaches.
- Developed and supported dashboards, GIS applications, recurring reporting products, risk analyses, maps, and executive decision-support tools integrating program and geographic information.
- Established SOPs, metadata standards, data-sharing approaches, governance practices, and training that strengthened information quality, consistency, and institutional use.
- Integrated program, operational, geographic, and contextual datasets into reusable analytical and decision-support workflows.
- Evaluated emerging GIS, data, and digital technologies and developed proof-of-concept applications to assess feasibility and operational value before broader adoption.
- Translated requirements from multiple technical offices into common GIS, data, and information-management solutions.
- Provided technical support, documentation, and knowledge transfer that strengthened internal capability to use and sustain GIS and analytical services.

## **GIS / Data Specialist**

*USAID / Credence Management Solutions, Office of HIV/AIDS | Washington, DC / Multi-Country | Nov 2020 - May 2021*

- Provided GIS, data, and analytical technical support across eight USAID Missions, supporting HIV surveillance, program monitoring, resource targeting, and country-level decision-making.
- Developed maps, dashboards, scenario models, and analytical products used to identify service gaps and prioritize follow-up.
- Integrated program, geographic, and operational datasets using Python, SQL, ArcGIS, and ETL workflows.
- Automated recurring analytical and data-processing tasks, improving consistency and turnaround for distributed country teams.
- Troubleshoot data, automation, analytical, and GIS workflow issues and supported technical users across multiple operating environments.
- Translated program and operational requirements into reusable analytical and geospatial solutions.

## **GIS Solution Engineer**

*Geospatial Consulting Group International / U.S. Census Bureau | Alexandria, Virginia | Mar 2020 - Nov 2020*

- Designed enterprise GIS solutions for federal-government requirements spanning ArcGIS Enterprise, cloud environments, databases, web applications, dashboards, Python automation, and SQL.
- Gathered business and technical requirements and translated them into scalable GIS architecture, data models, applications, and implementation approaches.
- Supported integration among GIS applications, enterprise datasets, APIs, databases, analytical tools, and automated processing workflows.
- Developed Python and SQL automation supporting repeatable data processing and enterprise GIS operations.
- Supported technical evaluation and troubleshooting of GIS applications, data services, database workflows, and system configurations.
- Worked across multidisciplinary technical teams to coordinate GIS requirements with application, cloud, database, and infrastructure environments.

## **GIS Technical Lead**

*DC Water and Sewer Authority | Washington, DC | Sep 2019 - Mar 2020*

- Led enterprise GIS and data support within a major water and wastewater utility, coordinating technical requirements across IT, engineering, field operations, and operational teams.
- Supported business-critical GIS applications, enterprise spatial data, web services, and information workflows used by engineering and field personnel.
- Provided technical leadership for GIS platform, application, database, and data issues and coordinated troubleshooting across technical and operational teams.
- Supported enterprise geospatial data-management and multi-user editing workflows to maintain reliable infrastructure information.
- Worked with engineering and operational stakeholders to translate business requirements into GIS data, application, and workflow improvements.
- Supported maintenance and evolution of corporate GIS capabilities within an operational utility environment where system reliability and authoritative data were important to daily operations.
- Provided technical support and knowledge transfer to users working with enterprise GIS tools and operational workflows.

## **Founder & Chief Executive Officer**

*eAdrasha | Technology Start-Up | Addis Ababa, Ethiopia | 2017 - 2020*

- Founded and led a location-technology company, taking a digital product from concept through requirements, development, testing, deployment, user feedback, and continuous improvement.
- Worked directly with software developers to translate customer and operational needs into product requirements, prioritized features, application workflows, and technical capabilities.

- Built and managed a 40+ person multidisciplinary organization across software development, data, field operations, quality assurance, partnerships, customer engagement, and product delivery.
- Reached more than 25,000 users within six months and developed partnerships with government organizations, businesses, transport companies, delivery providers, and other institutions.
- Developed capabilities supporting business search, digital addressing, routing, geocoding, APIs, and reusable location-data services.
- Established feedback loops between users, field teams, customer-facing staff, and software developers to improve functionality and product usability.

### Senior Geospatial Analyst

*Prince George's County Government, Office of Information Technology | Largo, Maryland | May 2014 - Sep 2019*

- Provided technical leadership for countywide enterprise GIS services supporting multiple government departments, operational functions, and public-facing applications.
- Led GIS strategy, enterprise data governance, application delivery, staff supervision, production support, and cross-agency geospatial standards.
- Supported enterprise GIS applications and data used by public works, emergency management, planning, licensing, infrastructure, and citizen-service programs.
- Coordinated GIS, database, application, and operational requirements across central IT and county departments.
- Supported multi-user enterprise geospatial data environments and governed data-maintenance workflows used across county government.
- Troubleshot GIS application, database, data, and workflow issues and coordinated resolution with technical teams and business users.
- Established standards and governance practices that improved GIS data quality, consistency, maintainability, and enterprise use.
- Led and supported staff responsible for GIS analysis, application delivery, data management, and departmental technical support.
- Served as a technical liaison between enterprise IT and operational departments, translating business requirements into sustainable GIS solutions.

### Geospatial Consultant

*International Development Assignments | Haiti, Malawi & Other Assignments | 2012 - 2017*

- Delivered geospatial analysis, digital mapping, information management, data governance, and capacity strengthening for international development and public-health programs.
- Designed GIS workflows and analytical products appropriate to country-level implementation environments and available institutional capacity.
- Worked with government and program stakeholders to define requirements, strengthen data-management processes, and improve use of geographic information.
- Developed documentation, standards, training, and knowledge-transfer materials supporting local ownership of geospatial systems and processes.

## EDUCATION & PROFESSIONAL CREDENTIALS

Master of Science, Geospatial Information Sciences | University of Maryland, College Park

Bachelor of Science, Environmental Science | University of Maryland, College Park

**Certified GIS Professional (GISP)**

## ADDITIONAL INFORMATION

**Regional Experience:** Ethiopia • Burundi • Haiti • Malawi • Multi-country USAID programs • United States

**Sector Experience:** Public Sector • International Development • Digital Health • Public Utilities • Water & Wastewater • Electricity Utility Advisory • Infrastructure • Local Government • Technology Entrepreneurship

**Languages:** English • Amharic

**Citizenship:** U.S. Citizen